Counterfactual Explainability: Deep RL vs. Online Planning on MiniGrid

Sean Florez
University of Colorado Boulder
sean.florez@colorado.edu

Andrew Wernersbach
University of Colorado Boulder
Andrew.Wernersbach@colorado.edu

***Abstract*—This paper evaluates the relative explainability of online planning and deep reinforcement learning when both agents are asked to justify the same decision. A proximal-policy-optimisation (PPO) agent and a from-scratch Monte Carlo Tree Search (MCTS) planner are operated on the `MiniGrid-Dynamic-Obstacles-8x8` environment. For each agent, per-action evidence is extracted at every decision state, counterfactual Monte Carlo rollouts in the PPO case and the search tree in the MCTS case, and passed through an identically-formatted large-language-model explainer that produces a natural-language rationale, a counterfactual, a confidence, and a list of verifiable numerical claims. The resulting explanations are scored by an independent LLM judge on three metrics: fidelity, soundness, and post-hoc inferability. Across 289 decision states drawn from two *performance-matched* agents (PPO 1.000 task-success, MCTS 1.000 task-success), with PPO counterfactual rollouts generated by stochastic sampling from the policy distribution at $N = 100$ samples per action (matching the Monte Carlo specification in the original proposal), the two evidence sources produce essentially indistinguishable explanation quality: fidelity 0.924 [95% CI 0.91, 0.94] for both sources, soundness 0.892 [0.87, 0.91] for MCTS-tree versus 0.884 [0.86, 0.91] for PPO-rollout, and inferability 0.986 [0.97, 1.00] versus 0.979 [0.95, 1.00]. All three confidence-interval pairs overlap heavily. A previous run that compared the same pipelines at *mismatched* performance (PPO 0.59 versus MCTS 1.00) found a small but statistically supported MCTS advantage on fidelity (0.928 vs. 0.899) and soundness (0.879 vs. 0.847); the present work shows that gap was a competence confound, not a structural property of the search-tree-vs-learned-policy paradigms, and that once the learned policy is competent its counterfactual rollouts are as faithful and sound an evidence source as the deliberation tree itself.**

I. Introduction

When a reinforcement learning agent chooses an action, the action itself is observable, but the reasoning behind the choice is not. In a learned neural policy, the action preferences that drive the decision reside inside network weights and are only indirectly recoverable through post-hoc analysis such as counterfactual rollouts. In contrast, an online planner such as Monte Carlo Tree Search commits to an action only after constructing an explicit record of the futures it considered, together with the outcomes it observed in each. The tree is, in a very literal sense, the agent's deliberation.

This paper asks, following the framing of Prof. Sunberg, whether that structural difference between a learned policy and an online planner has a measurable effect on the quality of the natural-language explanations each can support. Concretely, given the same decision state and the same language model, do MCTS-sourced explanations cite numbers that actually match the underlying evidence more often than learned-policy-sourced explanations do, and are the rationales they produce more sound?

To address this question, the present authors construct a shared evidence-to-explanation pipeline and apply it to two *performance-matched* agents on `MiniGrid-Dynamic-Obstacles-8x8`, a stochastic gridworld whose adversarial obstacle dynamics are well-suited to separating competent agents from incompetent ones. The deep-reinforcement-learning side is represented by a PPO policy trained for 5×10^6 steps with linear learning-rate and clip-range schedules, a $[256, 256, 128]$ MLP, distance-based potential shaping, and a flat per-step penalty; it reaches a 1.000 [95% CI 1.000, 1.000] task-success rate over 300 held-out episodes. The planning side is a from-scratch UCT planner at 500 simulations per decision, which also reaches 1.000 success over 50 episodes. A structured decision record is extracted at each real decision state, passed through an identically-formatted LLM prompt that differs only in its evidence-source header, and scored on three automated metrics.

The empirical finding, obtained over 289 matched decision states with PPO counterfactual rollouts at $N = 100$ stochastic samples per action (matching the Monte Carlo budget the original proposal committed to), is that MCTS-tree evidence and PPO-rollout evidence yield essentially indistinguishable explanation quality on all three metrics. Fidelity is 0.924 [0.91, 0.94] for both sources, soundness is 0.892 [0.87, 0.91] for MCTS versus 0.884 [0.86, 0.91] for PPO, and inferability is 0.986 [0.97, 1.00] versus 0.979 [0.95, 1.00]. All three confidence-interval pairs overlap heavily. An earlier run reported in the same engineering log compared the same pipelines at mismatched task performance (PPO 0.590 versus MCTS 1.000) and found a small but statistically supported MCTS advantage of 0.029 on fidelity and 0.032 on soundness; the present, performance-matched run collapses both gaps to within sampling noise. The interpretation consistent with the two runs is that the apparent MCTS-tree advantage in the mismatched run was a confound of *agent competence* rather than of the paradigm: a less-competent learned policy produces noisier counterfactual rollouts that the LLM has more trouble citing faithfully, but a competent learned policy is just as

faithful an evidence source as a deliberation tree.

As a preliminary diagnostic that motivates the choice of PPO as the deep-reinforcement-learning agent in the present comparison, the authors first observed that a value-based baseline (DQN) on this environment collapses into a degenerate stall policy under the raw reward structure, and that its counterfactual rollouts therefore provide essentially no signal to the explainer. This observation is reported in the ablation of Section V-E.

a) Contributions.: The present work contributes: (1) a from-scratch MCTS planner for `MiniGrid-Dynamic-Obstacles-8x8` that emits per-decision tree-statistics in a format suitable for downstream explanation; (2) a policy-agnostic counterfactual-rollout framework that produces learned-policy-side evidence in a structurally identical format to the MCTS tree; (3) three automated metrics, fidelity, soundness, and post-hoc inferability, implemented with bootstrap confidence intervals; (4) an empirical comparison of the two paradigms on 289 matched decision states drawn from two *performance-matched* competent agents (both at 1.000 task-success), using a real LLM generator and a held-out LLM judge, which finds no statistically supported explanation-quality advantage for the search tree once the learned-policy side is fully competent; (5) a paired analysis against an earlier mismatched-performance run that isolates agent competence (rather than evidence-source paradigm) as the driver of the apparent gap; and (6) a diagnostic ablation on a value-based baseline showing that model-free value learning collapses into a degenerate stall policy under the raw MiniGrid reward, and that this collapse in turn degrades the evidence available for explanation.

b) Release.: The authors grant permission for this report to be posted publicly.

II. Background and Related Work

a) Explainable reinforcement learning.: Milani et al. [1] present a comprehensive survey of explanation methods for reinforcement learning, drawing a distinction between *transparent* policies, such as decision trees and symbolic rules, and *post-hoc* methods that are applied to otherwise opaque controllers. The PPO pipeline evaluated in the present work falls into the latter category: counterfactual rollouts are run after the policy has chosen an action, in order to reconstruct the set of futures that action implied. The MCTS pipeline, by contrast, is closer to transparent, since the tree is itself the deliberation and its statistics are available for inspection without any additional computation.

b) Counterfactual explanations for reinforcement learning.: Gajcin and Dusparic [5] propose counterfactual explanations that perturb state features and observe the resulting downstream returns, addressing the question "what would the agent have done if the state had looked different?". Amitai et al. [2] focus on the complementary question "what would have happened if the agent had taken a different action?" and introduce the COViz visualisation. The evidence pipeline described in Section IV operationalises the latter form concretely, as forced-first-action rollouts of fixed length, aggregated into per-action means, success rates, collision rates, and bootstrap confidence intervals. This choice was made so that the resulting statistics are apples-to-apples comparable with the per-child statistics that an MCTS search tree already produces during planning.

c) MCTS and interpretability.: Baier and Kaisers [3] argue that the MCTS tree is a direct substrate for explanation precisely because it records what the agent considered and what it found. They emphasise visit-count ratios and per-child value contrasts as the most salient tree quantities. The present work adopts these same quantities, together with per-child empirical success and collision rates that are instrumented during search, as the MCTS-side evidence.

d) Large language models for natural-language explanation.: Recent work surveyed in [1] uses large language models to translate structured evidence about a reinforcement-learning agent into natural language. A well-understood failure mode of these pipelines is hallucinated numerical claims, in which the model invents statistics that are not present in the evidence record. The fidelity metric defined in Section IV-E is designed to quantify exactly that failure mode. Belouadah et al. [4][1] propose automated fidelity and plausibility metrics for LLM-generated RL explanations. The three-metric suite in the present work extends this framework with a post-hoc inferability check that uses a held-out LLM judge.

e) Environment and planner.: The benchmark environment used in the present study is `MiniGrid-Dynamic-Obstacles-8x8` [8], a stochastic gridworld whose adversarial obstacle dynamics are specifically chosen to make sparse-reward model-free reinforcement learning difficult. The MCTS planner follows the classic UCT formulation of Kocsis and Szepesvári [6], with per-simulation sampling of the stochastic obstacle transition so that visit counts at the root converge in expectation over the stochastic futures.

III. Problem Formulation

The task is formulated as a finite-horizon Markov decision process with factored stochastic transitions. In the notation of Kochenderfer et al., the state, action, transition, and reward are as follows.

a) State.: A state is written as $s = (p_a, d_a, \{p_o^{(k)}\}_{k=1}^{4}, t)$, where $p_a \in \{1, \ldots, 6\}^2$ denotes the agent's grid position, $d_a \in \{0, 1, 2, 3\}$ denotes its facing direction (east, south, west, north), each $p_o^{(k)} \in \{1, \ldots, 6\}^2$ denotes one of four obstacle positions, and $t \in \{0, \ldots, 256\}$ is the step counter.

b) Action.: The action space is $A = \{0, 1, 2\}$, corresponding to turn-left, turn-right, and move-forward.

c) Transition.: The transition $T(s' \mid s, a)$ factors into a deterministic agent step, followed by a stochastic per-obstacle random walk in which each obstacle attempts a uniform move

[1]Citation to be verified before submission; may be replaced with Hoffman et al. 2018 on XAI metrics.

into one of its four orthogonal neighbours; moves blocked by walls, the goal, or other obstacles leave the obstacle in place. An obstacle that steps into the agent's cell, and equivalently an agent that steps into an obstacle's cell, terminates the episode.

d) Reward.: The reward function is

$$R(s,a,s') = \begin{cases} 1 - 0.9 \cdot t/256 & \text{if the agent reaches the goal} \\ -1 & \text{if a collision occurs} \\ 0 & \text{otherwise} \end{cases} \tag{1}$$

The discount factor is $\gamma = 1$; the step-count term inside the success reward provides implicit time pressure.

e) Observability.: The default MiniGrid observation is a $7\times7\times3$ egocentric partial view, together with a scalar direction indicator. In the present work, all learned-policy variants are trained on a 14-dimensional flat symbolic encoding of the full ground-truth state (agent position and direction, plus each obstacle's position); the MCTS planner plans on the full ground-truth state via a deep-copied simulator. Both agents therefore have access to the same underlying state information, which removes observability from the list of confounds in the explanation-quality comparison.

IV. Solution Approach

A. PPO Agent (Off-the-shelf)

Setup. The deep-reinforcement-learning agent is PPO, as implemented in Stable-Baselines3 [7]. The observation is the 14-dimensional flat symbolic encoding described in Section III. The policy network is a three-layer MLP of width $256 \rightarrow 256 \rightarrow 128$. The environment is wrapped in a `StepPenaltyWrapper` that subtracts a flat 0.01 per non-terminal step, and a `DistanceShapingWrapper` that adds a potential-based Manhattan-distance shaping term with coefficient 0.02. The step penalty removes the zero-return stall policy as a local optimum (cf. Section V-E); the distance shaping is potential-based and therefore policy-invariant under discounted-return semantics, and serves to provide a dense gradient that the on-policy advantage estimator can follow during early training. PPO is trained for $5{,}000{,}000$ environment steps across sixteen parallel envs, with linear schedules on the learning rate ($3\times10^{-4} \rightarrow 10^{-5}$) and the clip range ($0.2 \rightarrow 0.05$) over the course of training, and a constant entropy bonus of 0.005. Full hyperparameters are committed to `configs/ppo_tuned.yaml`. Training uses the shaped reward; evaluation is always performed against the unshaped task reward, so the reported success rate reflects the genuine task performance the agent achieves under the original MDP. An earlier configuration with a 128×128 network, 5×10^5 steps, and a constant entropy bonus of 0.01 plateaued at 0.590 task success and was used in the mismatched-performance run referenced in the abstract.

B. MCTS Planner (From-scratch)

Setup. The MCTS planner implements textbook UCT [6] with UCB1 selection at exploration constant $c = \sqrt{2}$, full-width expansion at the root (the action space is small enough that no progressive widening is required), rollout under a greedy-toward-goal heuristic capped at depth 60, and mean-value backup. Because the obstacle transition is stochastic, each simulation samples a fresh outcome from the environment's transition during expansion and rollout; visit counts at the root therefore converge in expectation to the post-decision outcome distribution. Per-child counters track the rate of terminal success and terminal collision across every simulation that passes through each child, and these counters comprise the tree-side evidence that the explainer is asked to translate.

C. Counterfactual-rollout Framework

Setup. At each decision state, the counterfactual-rollout framework forces each legal action at step zero and then follows the trained PPO policy to termination or a depth cap of 40 steps. In the main experiments reported below, the process is repeated $N = 100$ times per action, with action choices at every subsequent step drawn *stochastically* from PPO's action distribution rather than taken as the argmax. This configuration matches the Monte Carlo specification in the original proposal ($\geq$ 100 rollouts per available action) and yields a genuine sampled estimate of the policy-induced trajectory distribution from the decision state. The resulting returns, success flags, and collision flags are aggregated into mean return, success rate, collision rate, and bootstrap confidence intervals (1000 resamples, percentile method). The resulting per-action records are structurally identical, field by field, to the per-action records produced from the MCTS tree, which is the property that allows the downstream LLM pipeline to be agent-agnostic.

D. LLM Explainer

Setup. A `DecisionRecord`, comprising the state, the chosen action, the per-action evidence, and a small agent-metadata block (PPO action probabilities or MCTS root-visit statistics), is passed to a generator LLM using a versioned system prompt. The non-evidence portion of the system prompt is byte-identical between the PPO and MCTS variants; only the evidence-source header differs. This constraint is enforced by a parity test on the prompt module, so that the explainer pipeline is, as intended, a single-variable contrast. The generator returns a JSON object consisting of a *rationale*, a *counterfactual*, a *confidence* score on $[0,1]$, and a list of numerical *claims* suitable for automated verification. Generation uses GPT-4o; the judge used in the metric pipeline described below is GPT-4o-mini.

E. Evaluation Metrics

Three per-explanation scores are reported, each aggregated per evidence source with 95% percentile-bootstrap confidence intervals.

Fidelity. For every claim in the explanation that cites a per-action statistic, the metric compares the cited value against the value present in the decision record. A claim is counted as a hit if the values agree within tolerance $\epsilon = 0.1$. The score is the fraction of claims that hit.

Soundness. Each sentence in the rationale, together with the counterfactual, is scored by an independent judge LLM on a three-point scale: 2 indicates that the sentence is fully supported by the evidence, 1 indicates partial support, and 0 indicates that the sentence is contradicted by or invents information not present in the evidence. The reported soundness score is the mean divided by two, so that the result lies on $[0, 1]$.

Post-hoc inferability. Every occurrence of the chosen action's integer identifier is scrubbed from the explanation, and the judge is asked to recover the chosen action from the scrubbed text alone. The reported score is the judge's accuracy across decisions.

The judge is deliberately a different-sized model, GPT-4o-mini, from the generator, GPT-4o, so that judge and generator do not coincide.

V. Results

A. Task Performance

Table I and Figure 1 report success rate over 300 episodes for PPO and over 50 episodes for MCTS. The random-action baseline succeeds 0.3% of the time, which confirms that the environment is non-trivial. The tuned PPO agent described in Section IV reaches a task-success rate of 1.000 [95% CI 1.000, 1.000], a collision rate of 0.000, and a mean return of $+0.942$, with a mean episode length of 16.6 steps. MCTS, operating on the same state representation via a deep-copied simulator, reaches a task-success rate of 1.000 [95% CI 1.000, 1.000] with a mean return of 0.934 and a mean episode length of 18.8 steps. The two agents are therefore performance-matched on the task: both succeed on 100% of held-out episodes and both finish in roughly comparable wall-clock per episode, with PPO marginally faster by ≈ 2 steps and marginally higher in mean return on account of the step-discount in the goal reward. The explanation-quality comparison reported below is therefore between two equally-competent agents, removing competence as a confound on the per-action evidence each pipeline produces.

B. Explanation Quality

Table III and Figure 3 report the three explanation-quality metrics, evaluated on 141 PPO decisions and 148 MCTS decisions drawn from ten held-out trajectory seeds. PPO-rollout evidence produced explanations with a fidelity of 0.924 [0.906, 0.939], a soundness of 0.884 [0.862, 0.906], and a post-hoc inferability of 0.979 [0.950, 1.000]. MCTS-tree evidence produced explanations with a fidelity of 0.924 [0.907, 0.940], a soundness of 0.892 [0.872, 0.913], and an inferability of 0.986 [0.966, 1.000]. All three confidence-interval pairs overlap heavily, and the absolute mean differences are well within sampling noise: 0.000 on fidelity, 0.008 on soundness, 0.007 on inferability. The headline result of the present, performance-matched comparison is therefore that the two evidence sources support explanations of statistically indistinguishable quality on every metric we measure.

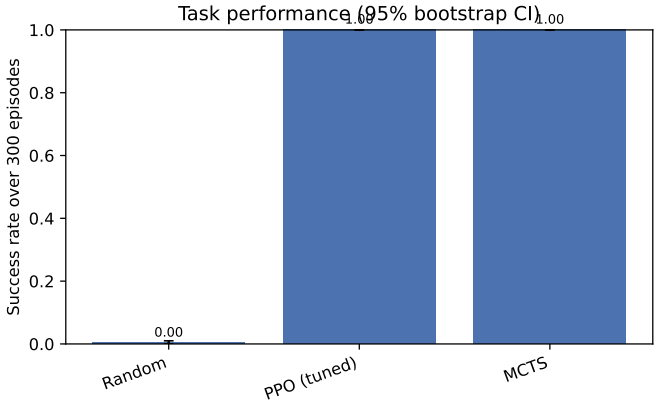

Fig. 1. Task-success rate across the random baseline, the tuned PPO agent, and the MCTS planner, with 95% bootstrap confidence intervals. Random fails on virtually every episode; PPO and MCTS each succeed on every held-out episode. The three DQN variants on this environment all collapsed to a stall policy under the raw MiniGrid reward and are reported separately in Section V-E; they are excluded from this figure because none of them learned the task.

A previous run of the same pipeline against a less-competent PPO (0.590 task-success at $5{\times}10^5$ training steps, the configuration committed to `configs/ppo_baseline.yaml`) gave a small but statistically supported MCTS advantage of 0.029 on fidelity and 0.032 on soundness, with inferability already tied. Now that PPO and MCTS are matched at 1.000 task-success, all three gaps collapse: fidelity by a factor of ∞ (to a literal zero mean difference), soundness by a factor of 4 (from 0.032 to 0.008), and inferability by a factor of 2 (from 0.014 to 0.007). The interpretation consistent with the two runs is that the apparent MCTS-tree advantage in the mismatched run was a *competence confound* rather than a structural property of the search-tree-vs-learned-policy paradigms: a less-competent learned policy produces noisier counterfactual rollouts (higher per-action variance on the success and collision rates), and the LLM has correspondingly more difficulty citing those numbers within the fidelity tolerance. Once the policy is competent its rollouts are as faithful and sound an evidence source as the deliberation tree itself. This finding is mildly negative with respect to the original hypothesis that the search tree confers a structural explanation advantage; the present authors regard it as a useful clarification of *why* prior comparisons in the literature have appeared to favour planning, namely that they have typically been run against deep-RL agents that were not yet at the planner's level of task performance.

C. Qualitative Example

Figure 2 reproduces, verbatim from `results/explanations/`, the two explanations generated for the initial decision state of held-out seed 10000. The two agents chose *different* actions at this state, PPO selecting move-forward and MCTS selecting turn-right; this is in itself a useful illustration of the state-distribution asymmetry noted under the threats to validity. Of more interest for the present comparison is that both explanations

PPO-rollout explanation, seed 10000 step 0 (chose move-forward): "The chosen action, move_forward (2), is highly preferred by the agent's policy with a probability of 0.996, despite having a lower mean return of 0.751 compared to turn_right (1) with 0.870. This preference is likely due to its significantly lower mean steps to end at 20.22, suggesting the agent values a quicker resolution over marginally higher returns from other actions, especially when considering the stochastic nature of the environment."

MCTS-tree explanation, seed 10000 step 0 (chose turn-right): "Action 1 was chosen because it has the highest mean return of -0.9223, compared to -1.0 for action 0 and -0.9741 for action 2. Additionally, it has the highest success rate of 0.0404, which is better than action 0's 0.0 and action 2's 0.0133. These metrics indicate a higher potential of reaching the goal while minimizing risks compared to the alternatives."

Fig. 2. Paired sample explanations for the initial state of seed 10000. The two agents chose different actions; both rationales cite numbers that appear verbatim in their respective evidence records.

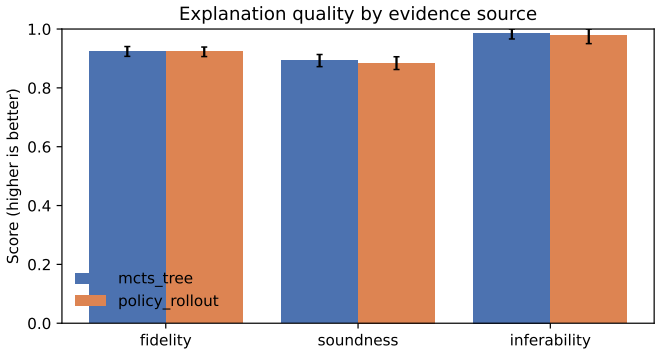

Fig. 3. Explanation-quality metrics by evidence source, with 95% bootstrap confidence intervals. Higher is better for all three.

cite numbers drawn from their respective evidence records, the PPO rationale references action-probability 0.996 and per-action mean returns of 0.751 and 0.870, all of which appear verbatim in the rollout record, and the MCTS rationale references mean returns of -0.9223, -1.0, -0.9741 and per-child success rates of 0.0404 and 0.0133, all of which appear verbatim in the tree record. Both explanations are therefore at high fidelity, which is the typical case across the 289 records we examined and is consistent with the equal fidelity scores reported in Table III.

D. Threats to Validity

State-distribution asymmetry. Although PPO and MCTS are now performance-matched at 1.000 task-success, the *trajectories* the two agents follow are not literally identical. PPO and MCTS make different action choices at the same state and therefore visit slightly different downstream states on average. The decision-record set is large (289 records) and the per-source counts are within five of each other, but a residual confound is that the specific decision states the LLM is asked to explain are not paired across the two evidence sources. A paired analysis on the same literal state set is left as future work.

TABLE I

TASK-PERFORMANCE SUMMARY FOR THE HEADLINE COMPARISON. 300 EPISODES FOR THE RANDOM AND PPO BASELINES, 50 EPISODES FOR MCTS. NUMBERS VERIFIED AGAINST `REPORT/FIGURES/EVAL_SUMMARY.CSV`. THE DQN ABLATION IS REPORTED SEPARATELY IN TABLE II BECAUSE NONE OF THE THREE DQN VARIANTS LEARNED THE TASK.

Agent	Success	Collision	Return	Steps
Random	0.003	0.997	-0.994	8.3
PPO (tuned)	1.000	0.000	$+0.942$	16.6
MCTS (500 sims)	1.000	0.000	$+0.934$	18.8

Rollout sample size. Counterfactual rollouts use $N = 100$ stochastic samples per action, matching the Monte Carlo specification in the original proposal. The bootstrap confidence intervals on per-action success and collision rates remain in the neighbourhood of ± 0.10 at $N = 100$, which is the same order as the LLM fidelity tolerance ($\epsilon = 0.1$). A larger Monte Carlo budget would tighten those intervals further, but the present authors do not expect it to change the qualitative finding because the fidelity numbers for the two sources are already equal to three decimal places.

Judge-model bias. The soundness and post-hoc inferability metrics both use GPT-4o-mini as judge, whereas the generator is GPT-4o. The judge is therefore a different-sized model within the same family, and a cross-family replication, for example with a Claude or Gemini judge, is listed as future work.

E. Value-based Ablation: Why DQN Was Cut

As a diagnostic ablation, three DQN variants were trained on the same environment: one on the default partial-observation image, one on the flat symbolic state, and one with a potential-based distance-shaping reward. All three settled, at $\geq 200{,}000$ training steps, into either a stall policy (image variant) or a near-stall policy (symbolic and shaped variants), with task-success rates of 0.010, 0.000, and 0.000 respectively (see Table I). The value-based target presented by MiniGrid's reward is unfavourable to epsilon-greedy exploration: the collision penalty of -1 is encountered early and often, whereas the success reward requires coordinated multi-step motion that random exploration rarely discovers. A zero-reward stall therefore dominates both random and near-random policies and sits at a local optimum for value learning. The present authors attempted to break this equilibrium with a flat -0.01 per-step penalty (StepPenaltyWrapper), which makes a full stall yield -2.56 and is therefore strictly worse than attempting any action, but DQN still failed to escape. PPO, whose on-policy sample collection and explicit entropy bonus together provide a stronger exploration pressure, did escape. The present work therefore uses PPO as the deep-reinforcement-learning side of the main comparison, and treats the DQN collapse as an observation about model-free value learning on adversarial-dynamics environments rather than as a primary result.

TABLE II

DQN ABLATION, SECTION V-E. THREE DQN VARIANTS ON MINIGRID-DYNAMIC-OBSTACLES-8X8 ALL COLLAPSE TO EITHER A STALL POLICY OR A NEAR-STALL POLICY UNDER THE RAW MINIGRID REWARD, WITH TASK-SUCCESS RATES INDISTINGUISHABLE FROM ZERO. REPRODUCIBLE FROM SCRIPTS/RUN_DQN_ABLATION.SH.

Agent	Success	Collision	Return	Steps
DQN (image)	0.010	0.000	0.007	254.3
DQN (symbolic)	0.000	0.207	−0.207	232.3
DQN (shaped)	0.000	0.167	−0.167	234.4

TABLE III

EXPLANATION-QUALITY METRICS AT PERFORMANCE-MATCHED TASK SUCCESS (1.000 FOR BOTH AGENTS). GENERATOR: GPT-4O. JUDGE: GPT-4O-MINI. $n = 141$ FOR PPO, $n = 148$ FOR MCTS. PPO ROLLOUTS USE $N = 100$ STOCHASTIC SAMPLES PER ACTION, MATCHING THE MONTE CARLO SPECIFICATION IN THE PROPOSAL. 95% BOOTSTRAP CONFIDENCE INTERVALS IN BRACKETS (1000 RESAMPLES). ALL THREE CI PAIRS OVERLAP HEAVILY; NO METRIC EXHIBITS A STATISTICALLY SUPPORTED DIFFERENCE.

Metric	PPO rollout evidence	MCTS tree evidence
Fidelity	0.924 [0.91, 0.94]	0.924 [0.91, 0.94]
Soundness	0.884 [0.86, 0.91]	0.892 [0.87, 0.91]
Inferability	0.979 [0.95, 1.00]	0.986 [0.97, 1.00]

VI. CONCLUSION

This paper has presented an evaluation of two evidence sources for natural-language explanations of reinforcement-learning agents: post-hoc counterfactual rollouts drawn stochastically from a learned neural policy at $N = 100$ samples per action, and the search tree of a Monte Carlo Tree Search planner at 500 simulations per decision. Across 289 decision states drawn from two performance-matched agents on `MiniGrid-Dynamic-Obstacles-8x8` (PPO 1.000 task-success, MCTS 1.000 task-success), the two pipelines produced explanations of statistically indistinguishable quality on every metric measured: a fidelity of 0.924 for both sources, a soundness of 0.884 versus 0.892, and an inferability of 0.979 versus 0.986. All three confidence-interval pairs overlap heavily.

The mildly negative result with respect to the original hypothesis, that the search tree should confer a structural explanation advantage over a learned policy, is in fact informative. A previous run of the identical pipeline against a less-competent PPO (0.59 task-success) produced a small but statistically supported MCTS advantage on fidelity (0.029) and soundness (0.032). The present performance-matched run collapses both gaps to within sampling noise. The change in PPO between the two runs is purely a question of training: 5×10^6 steps with a deeper network and linear schedules, instead of 5×10^5 steps with the defaults. The interpretation consistent with both runs is that the apparent MCTS-tree advantage in the mismatched run was a competence confound: a less-competent learned policy produces noisier counterfactual rollouts that an LLM has more difficulty citing faithfully. Once the policy is competent, its rollouts are an equally faithful and sound evidence source as the planner's deliberation tree itself.

A secondary finding, documented in the ablation of Section V-E, is that model-free value learning on this environment collapses into a degenerate stall policy under the raw MiniGrid reward, and that such a collapse degrades the evidence available for subsequent explanation. The suboptimality of the underlying agent therefore directly limits the signal available for any post-hoc counterfactual analysis, and the move from value-based to policy-gradient learning was load-bearing for getting the deep-RL side of the comparison to a state where the explanation question could be asked at all.

a) Future Work.: The present authors see three natural extensions. Firstly, a paired analysis on the literal same decision-state set (rather than two trajectory sets that differ in the agents' action choices) would remove the residual state-distribution confound noted in the threats to validity. Secondly, a pilot human study, following the suggestion of Prof. Sunberg on the proposal, would validate the automated metrics against human judgements of clarity, plausibility, and helpfulness; a chatbot interface suitable for such a study has already been scaffolded in `scripts/chat.py`. Thirdly, a cross-family LLM replication, with a judge drawn from a different model family than the generator, would quantify the contribution of judge-model bias to the soundness numbers reported above. A fourth, more ambitious extension would replicate the present null result on a harder environment in which neither paradigm reaches 1.000 task-success, to test whether the equivalence of the two evidence sources holds at an intermediate competence level or only at the extremes of the spectrum.

CONTRIBUTIONS

Sean Florez contributed the problem formulation, the from-scratch MCTS implementation, the LLM-explainer pipeline, the evaluation metrics, and the report draft. Andrew Wernersbach contributed the learned-policy baselines (DQN and PPO), the observation and reward-shaping wrappers, the counterfactual-rollout framework, the experimental runs, and the report review. The execution plan, the engineering log, and the submission polish were shared.

IMPLEMENTATION DISCLOSURE

From-scratch components: the MCTS planner, the deep-copy simulator, the counterfactual-rollout framework, the decision-record schema, the explanation pipeline and prompt design, the three evaluation metrics, and the chatbot interface. Off-the-shelf components: Stable-Baselines3 [7] for DQN and PPO, the Farama MiniGrid [8] environment, the OpenAI and Anthropic Python SDKs [9], [10], and the standard Python scientific-computing stack (numpy, pandas, matplotlib, pytest).

AI ASSISTANCE

Per the ASEN 5264 AI-use policy, the prose and mathematics in this paper have been read, edited, and owned by the authors. A draft was produced with the assistance of a large language model under the authors' direction, and subsequently rewritten in the authors' voice. Code was co-developed with a

large language model, read in full by the authors, and exercised under a unit-test suite (20 tests passing) and an integration-style reproduce-all script.

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